STA304: Guest Lecture
– Nov.18, 2025

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Today’s Plan

- Introduction & Background
- Research & Reading Courses
- Graduate School Application
- Graduate School Experience…so far :)

Intro!

My Journey at UofT

- 2021 – 2025
  - BSc. with high distinction
    - Data Science
    - Statistics
    - Economics

- 2025 – 2027
  - MEng. in Electrical and Computer Engineering (ECE)
    - Software Engineering
    - Deep Learning

Research!

Reading Course – STA496/STA497

- What is it?
  - 4 / 8 months research project === 0.5 credit
  - In the end:
    - 5 mins presentation
    - 2 – 3 mins Q & A

- Preparation:
  - Secure a supervisor
  - Draft a problem statement
    - Topic
    - Methodology
    - Evaluation Rubric

Estimating the association between multi-frequency noise measurements and sources of air pollution from oil and gas operations

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Background – Location

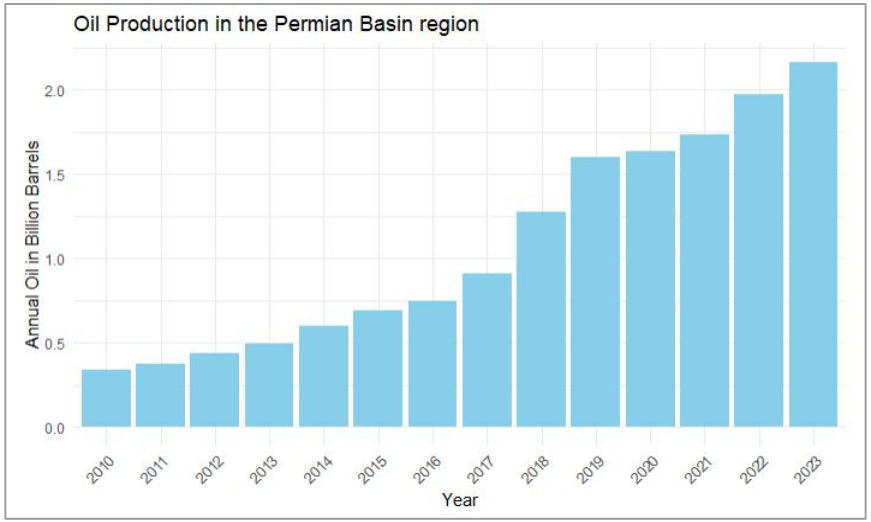
- The Permian Basin
  - straddles Texas and New Mexico
  - one of the most profitable and productive fracking locations in the world.





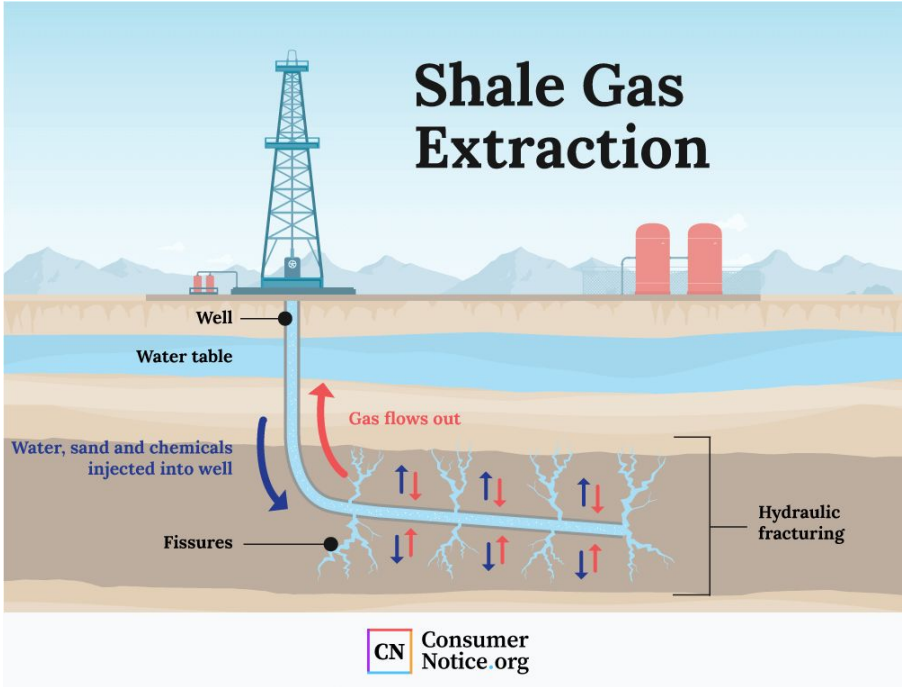
Background – Location

- Since 2010, the annual oil production has increased over 8-fold at wells located in the Permian Basin.
  → That's tons of fracking!



Background – Fracking

- Fracking
  - short for hydraulic fracturing
  - a method of extracting natural gas and oil from deep underground rock formations – shale
  - involves injecting high-pressure fluid into shale formations



Background – Fracking

- However, there are many environmental concerns around fracking
  - noise
  - air pollution (i.e. pm 2.5)
  - greenhouse gas emissions (methane and carbon dioxide)
  - radioactivity





Research Question

Research Question

- Question:
  - Estimating the association between frequencies and NOx

- Source Apportionment:
  - Find noise frequencies associated with NOx
    - Fracking process utilize different types of machine
    - Different machines have different noise frequency ranges
      → we try to find NOx related machines for regulation purposes

- NOx → Frequencies → Machine

Data Collection

Data – Collection

- 1-mins resolution data collected by a stationary monitoring platform, the “trailer”
  - sampling from May 1, 2023 to May 31, 2024
  - 30 noise frequencies from 12.5Hz to 16kHz, with 5 weighted frequencies
  - pollutants & greenhouse gasses: Nitrogen Oxides (NOx), Methane, etc





Data – Weighted Measures

- Weighted measures:
  - LAeq reflects human perception, weights in mid frequencies more.
  - LAFmax, LAFmin represent the max and min A-weighted fast response time (capture sub-minute sound pressure changes).
  - LC reflects weights low frequencies more, useful for engines, industrial operations.
  - LCpeak measures loudest peak in low frequency.

Data Cleaning

Data Cleaning

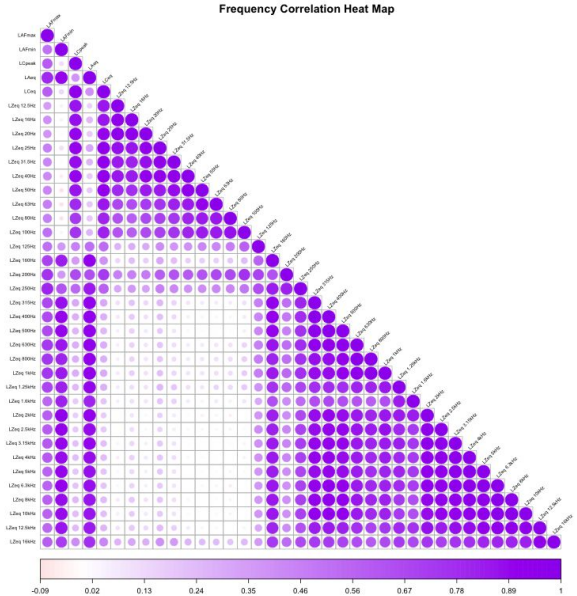
- Raw Data
  - 20+ folder of files with different formatting
  - 700+ datasets in each folder





Methodology

Noise Frequency I

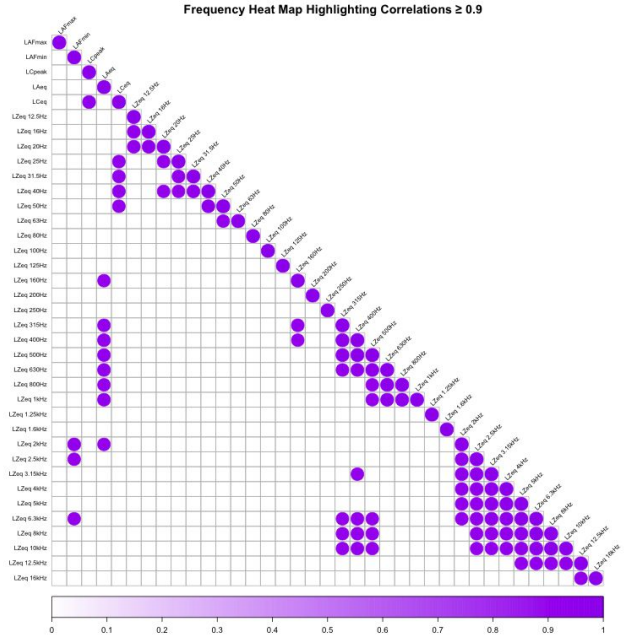


Correlation heatmap of all 30 frequencies and the 5 weighted measures

Correlation heatmap shows groupings of low and high frequencies.

→ highlight and groups where frequencies correlate > 0.9.

→ create differenced bands to help identify sources.



Correlations > 0.9

Noise Frequency II

1. Group the frequencies with high correlation (>0.9) into different bands
   a. Bands by color

2. Select the highest correlated frequency within the band as the representative (underlined in the band)

3. Select the frequencies that are not strongly correlated

Frequency	Frequency	Noise indicator	Difference Frequency	Metrological variable (station)
LZeq 12.5Hz	LZeq 500Hz	LAFmax	LCpeak - LCeq	Ambient Temp (F)
LZeq 16Hz	LZeq 630Hz	LAFmin	LAFmax - LAFmin	Wind Speed (m/s)
LZeq 20Hz	LZeq 800Hz	LCpeak	LZeq 25Hz -500Hz	Wind Direction (degrees)
LZeq 25Hz	LZeq 1kHz	LCeq	LZeq 250Hz -800Hz	Relative Humidity (%)
LZeq 31.5Hz	LZeq 1.25kHz	LAeq	LZeq 250Hz -500Hz	Pressure (Pa)
LZeq 40Hz	LZeq 1.6kHz		LZeq 500Hz -800Hz	Rainfall (mm)
LZeq 50Hz	LZeq 2kHz		LZeq 500Hz -5KHz	Solar Radiation (W/m^2)
LZeq 63Hz	LZeq 2.5kHz		LZeq 500Hz -12.5KHz	
LZeq 80Hz	LZeq 3.15kHz			Down/Up wind (derived)
LZeq 100Hz	LZeq 4kHz			Day/Night (derived)
LZeq 125Hz	LZeq 5kHz			
LZeq 160Hz	LZeq 6.3kHz			
LZeq 200Hz	LZeq 8kHz			
LZeq 250Hz	LZeq 10kHz			
LZeq 315Hz	LZeq 12.5kHz			
LZeq 400Hz	LZeq 16kHz			

Noise Frequency II

- Grey, blue, purple, and single lower-frequency potentially representing
  - compressors
  - generators
  - drilling

- Sounds can travel long distances and can cause vibrations.

Frequency	Frequency	Noise indicator	Difference Frequency	Metrological variable (station)
LZeq 12.5Hz	LZeq 500Hz	LAFmax	LCpeak - LCeq	Ambient Temp (F)
LZeq 16Hz	LZeq 630Hz	LAFmin	LAFmax - LAFmin	Wind Speed (m/s)
LZeq 20Hz	LZeq 800Hz	LCpeak	LZeq 25Hz -500Hz	Wind Direction (degrees)
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LZeq 250Hz	LZeq 10kHz			
LZeq 315Hz	LZeq 12.5kHz			
LZeq 400Hz	LZeq 16kHz			

Noise Frequency II

- Yellow and single mid-frequencies potentially representing:
  - pumps
  - machinery
  - pipelines

- Sounds are audible.

Frequency	Frequency	Noise indicator	Difference Frequency	Metrological variable (station)
LZeq 12.5Hz	LZeq 500Hz	LAFmax	LCpeak - LCeq	Ambient Temp (F)
LZeq 16Hz	LZeq 630Hz	LAFmin	LAFmax - LAFmin	Wind Speed (m/s)
LZeq 20Hz	LZeq 800Hz	LCpeak	LZeq 25Hz -500Hz	Wind Direction (degrees)
LZeq 25Hz	LZeq 1kHz	LCeq	LZeq 250Hz -800Hz	Relative Humidity (%)
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LZeq 40Hz	LZeq 1.6kHz		LZeq 500Hz -800Hz	Rainfall (mm)
LZeq 50Hz	LZeq 2kHz		LZeq 500Hz -5KHz	Solar Radiation (W/m²)
LZeq 63Hz	LZeq 2.5kHz		LZeq 500Hz -12.5KHz	
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LZeq 200Hz	LZeq 8kHz			
LZeq 250Hz	LZeq 10kHz			
LZeq 315Hz	LZeq 12.5kHz			
LZeq 400Hz	LZeq 16kHz			

Noise Frequency II

- Red and grey frequencies are high-frequency signatures potentially representing:
  - flaring
  - venting

- Sounds are audible and disturbing.

Frequency	Frequency	Noise indicator	Difference Frequency	Metrological variable (station)
LZeq 12.5Hz	LZeq 500Hz	LAFmax	LCpeak - LCeq	Ambient Temp (F)
LZeq 16Hz	LZeq 630Hz	LAFmin	LAFmax - LAFmin	Wind Speed (m/s)
LZeq 20Hz	LZeq 800Hz	LCpeak	LZeq 25Hz -500Hz	Wind Direction (degrees)
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Modelling – XGBoost

Boosting

- Building a strong model by combining many weak models
  - a weak model == a simple decision tree that is only slightly better than random
  - train one after another:
    - the later focuses on the mistake made by previous one
      → getting better step by step

- Learn from error → fix the error → repeat → better performance

- Overfitting?

Note: decision tree is a predictive model that makes decision by splitting data into small groups.

XGBoost

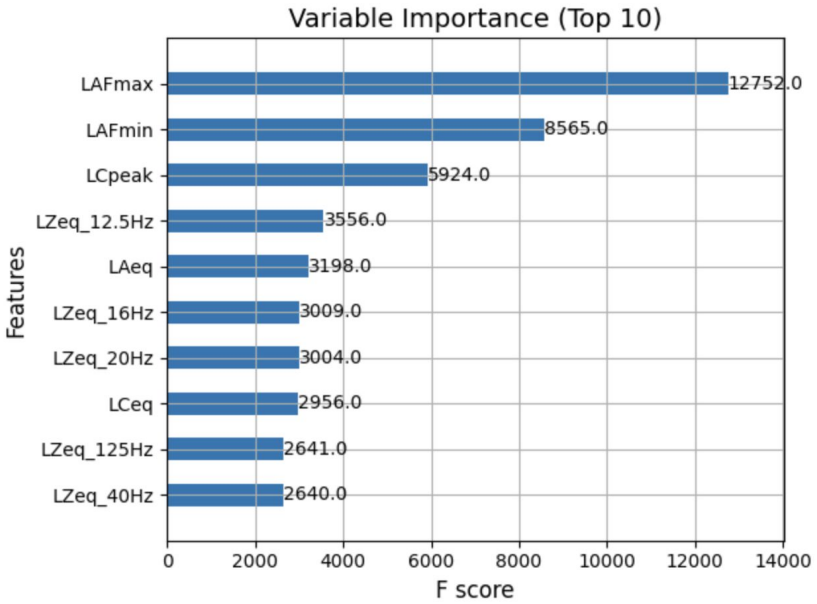
- eXtreme Gradient Boosting
  - eXtreme: extremely optimized
  - Gradient: gradient-based error == gradient of the loss

- Why XGBoost?
  - Regularization: prevents overfitting by penalizing complex trees
  - Tree Pruning: removes unnecessary brunches for efficiency
  - Handle Missing Value
  - Works well with spatial data

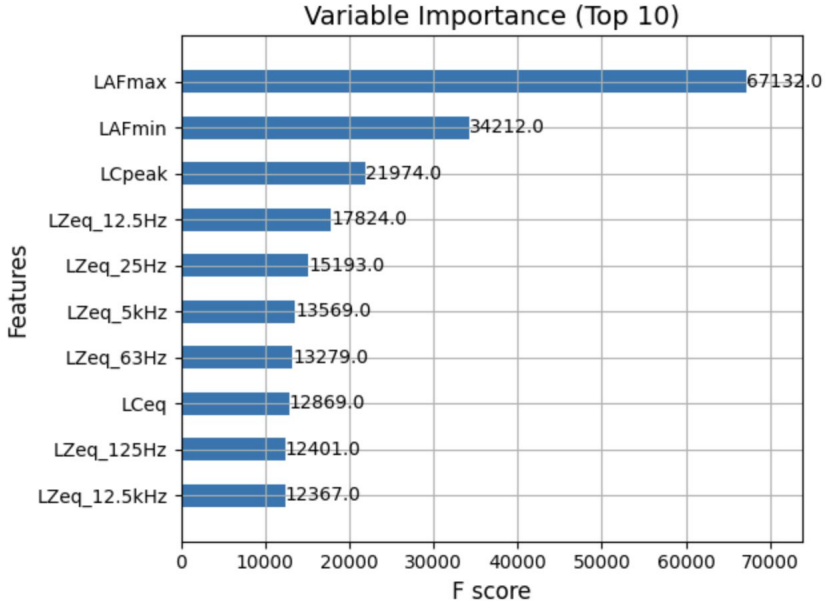
Preliminary Result

NOx Prediction: 2 XGBoost Models

1. Full Frequency Range
   - R-squared: 0.43



2. Selected Frequencies from grouping
   - R-Squared: 0.79



Any Questions?!

Grads School Apps!

Preparation

- Figure Out Direction / Programs
  - think ahead: change or continue?
  - consult experts

- Ask For Reference Letters
  - Mentor in intern / Instructor in class
  - Ask for it EARLY!!!

- Documents
  - CV / Resume
  - Transcript
  - Personal Statement
    === what and how is your passion? why program & school?

Grads School Feeling :)

A bit of Feeling

- Coursework
  - fewer courses, but heavier work
  - I can feel myself improving every single day !!

- Time Management
  - consistent progress
  - much better discipline, far less procrastination

- Personally
  - started writing
  - learning cooking
  - picked up old habits: swimming, city walks, exploring food spots





Thanks! Any Questions~